ADR Identification Basics: Case Analysis Report

Workora synthetic pharmacovigilance training pack | 18 fictional, de-identified cases (SYN-PV-001 to SYN-PV-018)

1. Objective and approach

This report identifies the suspect drug and the adverse drug reaction (ADR) in each case, correlates medication timelines with symptom onset, separates the suspect drug from co-medications, and records a causality category with its rationale and limitations. Every classification follows the pack rules: facts not present are kept **unknown**, seriousness is kept separate from severity, reporting frequency is not treated as evidence of causation, and the material is educational only.

Question	Criterion applied
Is the case valid?	Four minimum elements from the field guide: identifiable patient, identifiable reporter, suspect product, adverse event.
Suspect vs co-medication	Suspect = the product the reporter names as suspect, with the event arising after it began. Co-medications are listed as recorded; their start dates and doses are unknown, so they stay 'not excluded', not 'ruled out'.
Chronology	Onset is recorded as days after start of the suspect drug. Calendar dates, stop date and event duration are not supplied, so dechallenge timing cannot be verified.
Causality	WHO-UMC categories. 'Certain' and 'Probable/Likely' require that disease and other drugs are unlikely explanations, but medical history is unknown in every case, so the ceiling here is **Possible**. 'Conditional/Unclassified' is used when the record contradicts itself on dechallenge.
Seriousness	ICH E2A criteria (death, life-threatening, hospitalisation, disability, congenital anomaly, other medically important). This is separate from severity (intensity), which is **not supplied** for any case and is recorded as unknown.
Expectedness	Taken as supplied ('Known' / 'Unexpected'). The reference safety document is not named in the pack, so the value should be confirmed against the product label.

2. Key findings

- All **18** cases meet the four minimum validity elements (see Appendix A for the caveat on reporter identifiability).
- **8 suspect drug / reaction pairs** were identified: amoxicillin-rash, atorvastatin-myalgia, metformin-nausea, amlodipine-peripheral oedema, ciprofloxacin-tendon pain, sertraline-dizziness, omeprazole-headache, losartan-fatigue. Each case has exactly one suspect drug and one event.
- Causality: **13 Possible**, **5 Conditional/Unclassified**, none Certain/Probable (history unknown), none Unlikely. The Conditional cases are SYN-PV-003, 007, 008, 014 and 016, where the dechallenge field contradicts the narrative outcome.
- Seriousness as supplied: 5 hospitalisation, 10 medically important, 3 not serious. Severity is unknown in all cases. Only SYN-PV-008 and 016 are 'Unexpected', and both are non-serious.
- Outcome from the narratives: 6 recovered, 6 recovering, 6 not recovered. The suspect drug was stopped in all 18 cases.
- Data-quality problems matter more than the drug names: **15** of 18 indications do not match the suspect drug, **14** of 18 recorded doses fall outside the usual range (including three at 500 mg twice daily for amlodipine, omeprazole and atorvastatin), and **7** cases record a positive rechallenge with no documented reintroduction. These need source verification before any case could be relied on.

3. ADR identification summary table

Values are as supplied in case_scenarios.csv unless marked 'from narrative'. Reaction terms are verbatim, not MedDRA-coded. 'Unknown' means not provided.

Case	Age / Sex	Suspect drug, dose, route	Indication (as recorded)	ADR (reaction)	Onset (days)	Co-medication	Seriousness	Expected	Action / outcome (from narrative)	Reporter
SYN-PV-001	30 / Male	**Amoxicillin** 20 mg once daily, oral	Anxiety	**Rash**	17	None reported (not confirmed none)	Serious: medically important	Known	Stopped; not recovered	Consumer
SYN-PV-002	36 / Unknown	**Atorvastatin** 5 mg once daily, oral	Anxiety	**Myalgia**	1	None reported (not confirmed none)	Not serious	Known	Stopped; recovered	Pharmacist
SYN-PV-003	34 / Female	**Metformin** 5 mg once daily, oral	Anxiety	**Nausea**	13	Vitamin D	Serious: hospitalisation	Known	Stopped; recovered	Consumer
SYN-PV-004	61 / Male	**Amlodipine** 500 mg twice daily, oral	Anxiety	**Peripheral oedema**	21	None reported (not confirmed none)	Serious: medically important	Known	Stopped; recovered	Pharmacist
SYN-PV-005	33 / Unknown	**Ciprofloxacin** 500 mg twice daily, oral	Infection	**Tendon pain**	15	None reported (not confirmed none)	Serious: medically important	Known	Stopped; not recovered	Pharmacist
SYN-PV-006	31 / Unknown	**Sertraline** 20 mg once daily, oral	Hypertension	**Dizziness**	3	Paracetamol	Serious: medically important	Known	Stopped; recovering	Physician
SYN-PV-007	63 / Male	**Omeprazole** 500 mg twice daily, oral	Hypertension	**Headache**	16	Paracetamol	Serious: medically important	Known	Stopped; recovered	Consumer
SYN-PV-008	36 / Male	**Losartan** 5 mg once daily, oral	Infection	**Fatigue**	2	Paracetamol	Not serious	Unexpected	Stopped; recovering	Pharmacist
SYN-PV-009	78 / Unknown	**Amoxicillin** 5 mg once daily, oral	Anxiety	**Rash**	7	Vitamin D	Serious: medically important	Known	Stopped; recovering	Physician
SYN-PV-010	57 / Female	**Atorvastatin** 5 mg once daily, oral	Infection	**Myalgia**	21	Paracetamol	Serious: hospitalisation	Known	Stopped; recovered	Pharmacist
SYN-PV-011	23 / Unknown	**Metformin** 5 mg once daily, oral	Hypertension	**Nausea**	19	None reported (not confirmed none)	Serious: medically important	Known	Stopped; recovering	Consumer
SYN-PV-012	24 / Female	**Amlodipine** 5 mg once daily, oral	Hypertension	**Peripheral oedema**	4	None reported (not confirmed none)	Serious: medically important	Known	Stopped; not recovered	Pharmacist
SYN-PV-013	75 / Female	**Ciprofloxacin** 500 mg twice daily, oral	Diabetes	**Tendon pain**	5	Vitamin D	Serious: medically important	Known	Stopped; not recovered	Physician
SYN-PV-014	75 / Female	**Sertraline** 5 mg once daily, oral	Diabetes	**Dizziness**	9	Paracetamol	Serious: hospitalisation	Known	Stopped; recovering	Physician
SYN-PV-015	33 / Male	**Omeprazole** 20 mg once daily, oral	Hypertension	**Headache**	16	None reported (not confirmed none)	Serious: hospitalisation	Known	Stopped; recovered	Physician
SYN-PV-016	66 / Unknown	**Losartan** 20 mg once daily, oral	Diabetes	**Fatigue**	10	None reported (not confirmed none)	Not serious	Unexpected	Stopped; not recovered	Physician
SYN-PV-017	57 / Unknown	**Amoxicillin** 5 mg once daily, oral	Hypertension	**Rash**	21	Vitamin D	Serious: medically important	Known	Stopped; recovering	Physician
SYN-PV-018	56 / Female	**Atorvastatin** 500 mg twice daily, oral	Anxiety	**Myalgia**	17	None reported (not confirmed none)	Serious: hospitalisation	Known	Stopped; not recovered	Physician

4. Timeline correlation and causality assessment (WHO-UMC)

Time-to-onset comments rely on general pharmacological knowledge and should be checked against the product label. 'Undocumented' means the record gives no dates, dose or recurrence details for the dechallenge or rechallenge.

Case	Suspect / event	Time to onset	Dechallenge (field)	Rechallenge (field)	WHO-UMC	Rationale and limitations
SYN-PV-001	Amoxicillin Rash	Day 17: compatible (later end of typical range for delayed rash)	Negative (narrative: not recovered)	Positive (undocumented)	**Possible**	Rash recorded as Known; onset compatible. Negative dechallenge with no recovery (rash can persist after stopping). Positive rechallenge reported but undocumented. No concomitants reported; history unknown.
SYN-PV-002	Atorvastatin Myalgia	Day 1: atypically early for statin myalgia; not excluded	Unknown (narrative: recovered)	Negative (undocumented)	**Possible**	Myalgia recorded as Known, but day-1 onset is atypically early. Recovery after stopping is noted while the dechallenge field is Unknown. Negative rechallenge undocumented. Non-serious.
SYN-PV-003	Metformin Nausea	Day 13: compatible (GI effects usually earlier, but possible)	Negative (narrative: recovered)	Negative (undocumented)	**Conditional / Unclassified**	Nausea recorded as Known; onset compatible. Negative dechallenge conflicts with 'recovered' in the narrative, so the effect of withdrawal cannot be interpreted. Vitamin D: no relationship established; dose unknown.
SYN-PV-004	Amlodipine Peripheral oedema	Day 21: compatible	Positive (narrative: recovered)	Negative (undocumented)	**Possible**	Oedema recorded as Known; onset compatible. Positive dechallenge fits recovery, but a negative rechallenge is reported (undocumented), so evidence is mixed. Recorded dose is far above the usual range: verify; if true, dose-related effect or medication error is possible.
SYN-PV-005	Ciprofloxacin Tendon pain	Day 15: compatible	Unknown (narrative: not recovered)	Not done	**Possible**	Tendon pain recorded as Known; onset compatible; indication (infection) is consistent with the drug. Dechallenge unknown, not recovered, no rechallenge. No concomitants reported; other causes cannot be excluded as history is unknown.
SYN-PV-006	Sertraline Dizziness	Day 3: compatible	Positive (narrative: recovering)	Not done	**Possible**	Dizziness recorded as Known; onset compatible. Positive dechallenge, patient recovering. Paracetamol start date unknown (may have been taken for the symptom). Recorded indication (hypertension) does not match sertraline and could itself relate to dizziness.
SYN-PV-007	Omeprazole Headache	Day 16: compatible	Negative (narrative: recovered)	Negative (undocumented)	**Conditional / Unclassified**	Headache recorded as Known; onset compatible. Negative dechallenge conflicts with 'recovered'. Paracetamol may have been used to treat the headache (start date unknown). Dose unusually high; indication mismatch.
SYN-PV-008	Losartan Fatigue	Day 2: compatible (non-specific event)	Negative (narrative: recovering)	Not done	**Conditional / Unclassified**	Fatigue recorded as Unexpected; onset day 2. Negative dechallenge conflicts with 'recovering'. Recorded indication (infection) can itself cause fatigue; paracetamol co-medication. Non-serious, so unexpectedness alone does not meet expedited criteria.
SYN-PV-009	Amoxicillin Rash	Day 7: compatible	Positive (narrative: recovering)	Not done	**Possible**	Rash recorded as Known; day 7 is typical for delayed rash. Positive dechallenge, recovering. Age 78. Vitamin D: no recognised link on available data. Dose unusually low; indication mismatch.
SYN-PV-010	Atorvastatin Myalgia	Day 21: compatible	Unknown (narrative: recovered)	Negative (undocumented)	**Possible**	Myalgia recorded as Known; onset compatible. Recovered after stopping (dechallenge field Unknown). Paracetamol may be treating the myalgia; infection (recorded indication) can also cause myalgia. Negative rechallenge undocumented.
SYN-PV-011	Metformin Nausea	Day 19: compatible (later than typical)	Unknown (narrative: recovering)	Positive (undocumented)	**Possible**	Nausea recorded as Known; onset later than typical. Recovering; dechallenge field Unknown. Positive rechallenge undocumented. Dose unusually low; indication mismatch. No concomitants reported.
SYN-PV-012	Amlodipine Peripheral oedema	Day 4: early for oedema, but possible	Unknown (narrative: not recovered)	Positive (undocumented)	**Possible**	Oedema recorded as Known; onset early but possible. Indication (hypertension) and dose are consistent. Not recovered; dechallenge unknown. Positive rechallenge undocumented. No concomitants reported.
SYN-PV-013	Ciprofloxacin Tendon pain	Day 5: compatible (early onset described)	Negative (narrative: not recovered)	Positive (undocumented)	**Possible**	Tendon pain recorded as Known; age 75 is a recognised fluoroquinolone risk factor (general knowledge). Negative dechallenge and no recovery are consistent. Positive rechallenge undocumented. Indication mismatch (diabetes is itself a tendinopathy risk factor). Vitamin D: no known role.

Case	Suspect / event	Time to onset	Dechallenge (field)	Rechallenge (field)	WHO-UMC	Rationale and limitations
SYN-PV-014	Sertraline Dizziness	Day 9: compatible	Negative (narrative: recovering)	Negative (undocumented)	**Conditional / Unclassified**	Dizziness recorded as Known; onset compatible. Negative dechallenge conflicts with 'recovering'. Paracetamol co-medication; recorded diabetes (indication mismatch) could contribute. Dose unusually low. Hospitalisation details unknown.
SYN-PV-015	Omeprazole Headache	Day 16: compatible	Unknown (narrative: recovered)	Positive (undocumented)	**Possible**	Headache recorded as Known; onset compatible. Recovered after stopping (dechallenge field Unknown). Positive rechallenge undocumented. Hospitalisation reason/dates unknown. Recorded hypertension (indication mismatch) could itself cause headache.
SYN-PV-016	Losartan Fatigue	Day 10: compatible	Positive (narrative: not recovered)	Negative (undocumented)	**Conditional / Unclassified**	Fatigue recorded as Unexpected; onset day 10. Positive dechallenge conflicts with 'not recovered'. Negative rechallenge undocumented. Diabetes (recorded indication) can itself cause fatigue. Non-serious.
SYN-PV-017	Amoxicillin Rash	Day 21: compatible (later than typical)	Unknown (narrative: recovering)	Positive (undocumented)	**Possible**	Rash recorded as Known; onset later than typical. Recovering; dechallenge field Unknown. Positive rechallenge undocumented. Vitamin D: no recognised link. Dose unusually low; indication mismatch.
SYN-PV-018	Atorvastatin Myalgia	Day 17: compatible	Unknown (narrative: not recovered)	Positive (undocumented)	**Possible**	Myalgia recorded as Known; onset compatible. Not recovered; dechallenge unknown; positive rechallenge undocumented. Recorded dose is far above the usual range: possible entry error or overdose (unknown). Hospitalisation details unknown; indication mismatch.

Why no case reaches Probable: that category needs disease and other drugs to be unlikely explanations. Medical history is not supplied, concomitant start dates and doses are unknown, and several recorded indications do not match the suspect drug, so those alternatives cannot be judged. Cases move up only if follow-up supplies that information.

5. Suspect drugs versus co-medications

Case	Suspect drug (role: suspect)	Co-medication (role: concomitant, not suspect)	Assessment of the co-medication
SYN-PV-001	Amoxicillin	None reported	None reported by the reporter. This is not confirmation of no other medicines (over-the-counter, herbal, or unrecorded products stay unknown).
SYN-PV-002	Atorvastatin	None reported	None reported by the reporter. This is not confirmation of no other medicines (over-the-counter, herbal, or unrecorded products stay unknown).
SYN-PV-003	Metformin	Vitamin D	Start date and dose unknown. No relationship to the event is established from the information supplied; not excluded.
SYN-PV-004	Amlodipine	None reported	None reported by the reporter. This is not confirmation of no other medicines (over-the-counter, herbal, or unrecorded products stay unknown).
SYN-PV-005	Ciprofloxacin	None reported	None reported by the reporter. This is not confirmation of no other medicines (over-the-counter, herbal, or unrecorded products stay unknown).
SYN-PV-006	Sertraline	Paracetamol	Start date and dose unknown. May have been taken to treat the event (dizziness), so it could be a consequence rather than a cause. No role assigned.
SYN-PV-007	Omeprazole	Paracetamol	Start date and dose unknown. May have been taken to treat the event (headache), so it could be a consequence rather than a cause. No role assigned.
SYN-PV-008	Losartan	Paracetamol	Start date and dose unknown. May have been taken to treat the event (fatigue), so it could be a consequence rather than a cause. No role assigned.
SYN-PV-009	Amoxicillin	Vitamin D	Start date and dose unknown. No relationship to the event is established from the information supplied; not excluded.
SYN-PV-010	Atorvastatin	Paracetamol	Start date and dose unknown. May have been taken to treat the event (myalgia), so it could be a consequence rather than a cause. No role assigned.
SYN-PV-011	Metformin	None reported	None reported by the reporter. This is not confirmation of no other medicines (over-the-counter, herbal, or unrecorded products stay unknown).
SYN-PV-012	Amlodipine	None reported	None reported by the reporter. This is not confirmation of no other medicines (over-the-counter, herbal, or unrecorded products stay unknown).
SYN-PV-013	Ciprofloxacin	Vitamin D	Start date and dose unknown. No relationship to the event is established from the information supplied; not excluded.
SYN-PV-014	Sertraline	Paracetamol	Start date and dose unknown. May have been taken to treat the event (dizziness), so it could be a consequence rather than a cause. No role assigned.
SYN-PV-015	Omeprazole	None reported	None reported by the reporter. This is not confirmation of no other medicines (over-the-counter, herbal, or unrecorded products stay unknown).
SYN-PV-016	Losartan	None reported	None reported by the reporter. This is not confirmation of no other medicines (over-the-counter, herbal, or unrecorded products stay unknown).
SYN-PV-017	Amoxicillin	Vitamin D	Start date and dose unknown. No relationship to the event is established from the information supplied; not excluded.
SYN-PV-018	Atorvastatin	None reported	None reported by the reporter. This is not confirmation of no other medicines (over-the-counter, herbal, or unrecorded products stay unknown).

Drug-event pairs in the dataset

Suspect drug	Reaction	Cases	Case numbers (SYN-PV-)
Amlodipine	Peripheral oedema	2	004, 012
Amoxicillin	Rash	3	001, 009, 017
Atorvastatin	Myalgia	3	002, 010, 018
Ciprofloxacin	Tendon pain	2	005, 013
Losartan	Fatigue	2	008, 016
Metformin	Nausea	2	003, 011
Omeprazole	Headache	2	007, 015
Sertraline	Dizziness	2	006, 014

These counts only describe how many cases this training pack contains. They are not incidence rates, do not indicate a safety signal, and are not evidence of causation.

6. Data-quality checks and follow-up queries

Recorded values are kept as supplied; nothing has been corrected or imputed, and each flag is a query for the source, not a conclusion. Standard follow-up for every case: start/stop dates of the suspect drug, event onset and resolution dates, verified dose and units, medical history, co-medication start dates and doses, and whether the drug was ever reintroduced.

Case	Dose check (general reference ranges; verify with label)	Indication check	Internal conflicts / undocumented items	Seriousness note
SYN-PV-001	20 mg once daily: Unusually low (usual single doses are hundreds of mg)	Mismatch: Amoxicillin recorded for anxiety	Positive rechallenge recorded, but narrative says only that treatment was stopped	Reason for 'medically important' not stated; severity unknown.
SYN-PV-002	5 mg once daily: Below usual range (10-80 mg/day)	Mismatch: Atorvastatin recorded for anxiety	Dechallenge Unknown although narrative says 'recovered'	No seriousness criterion met as supplied; severity unknown.
SYN-PV-003	5 mg once daily: Unusually low (usual doses start at 500 mg)	Mismatch: Metformin recorded for anxiety	Dechallenge Negative vs narrative 'recovered'	Admission dates, reason and discharge status unknown.
SYN-PV-004	500 mg twice daily: Far above usual maximum (10 mg/day); entry error or overdose unknown	Mismatch: Amlodipine recorded for anxiety	None identified	Reason for 'medically important' not stated; severity unknown.
SYN-PV-005	500 mg twice daily: Typical	Consistent	None identified	Reason for 'medically important' not stated; severity unknown.
SYN-PV-006	20 mg once daily: Non-standard and below usual starting dose (25-50 mg)	Mismatch: Sertraline recorded for hypertension	None identified	Reason for 'medically important' not stated; severity unknown.
SYN-PV-007	500 mg twice daily: Far above usual range (20-40 mg/day); entry error or overdose unknown	Mismatch: Omeprazole recorded for hypertension	Dechallenge Negative vs narrative 'recovered'	Reason for 'medically important' not stated; severity unknown.
SYN-PV-008	5 mg once daily: Unusually low (usual 25-100 mg/day)	Mismatch: Losartan recorded for infection	Dechallenge Negative vs narrative 'recovering'	No seriousness criterion met as supplied; severity unknown.
SYN-PV-009	5 mg once daily: Unusually low (usual single doses are hundreds of mg)	Mismatch: Amoxicillin recorded for anxiety	None identified	Reason for 'medically important' not stated; severity unknown.
SYN-PV-010	5 mg once daily: Below usual range (10-80 mg/day)	Mismatch: Atorvastatin recorded for infection	Dechallenge Unknown although narrative says 'recovered'	Admission dates, reason and discharge status unknown.
SYN-PV-011	5 mg once daily: Unusually low (usual doses start at 500 mg)	Mismatch: Metformin recorded for hypertension	Dechallenge Unknown although narrative says 'recovering'; Positive rechallenge recorded, but narrative says only that treatment was stopped	Reason for 'medically important' not stated; severity unknown.
SYN-PV-012	5 mg once daily: Typical	Consistent	Positive rechallenge recorded, but narrative says only that treatment was stopped	Reason for 'medically important' not stated; severity unknown.
SYN-PV-013	500 mg twice daily: Typical	Mismatch: Ciprofloxacin recorded for diabetes	Positive rechallenge recorded, but narrative says only that treatment was stopped	Reason for 'medically important' not stated; severity unknown.
SYN-PV-014	5 mg once daily: Unusually low	Mismatch: Sertraline recorded for diabetes	Dechallenge Negative vs narrative 'recovering'	Admission dates, reason and discharge status unknown.
SYN-PV-015	20 mg once daily: Typical	Mismatch: Omeprazole recorded for hypertension	Dechallenge Unknown although narrative says 'recovered'; Positive rechallenge recorded, but narrative says only that treatment was stopped	Admission dates, reason and discharge status unknown.
SYN-PV-016	20 mg once daily: Non-standard, below usual starting dose (25-50 mg)	Consistent (partial: diabetic kidney indication only)	Dechallenge Positive vs narrative 'not recovered'	No seriousness criterion met as supplied; severity unknown.
SYN-PV-017	5 mg once daily: Unusually low (usual single doses are hundreds of mg)	Mismatch: Amoxicillin recorded for hypertension	Dechallenge Unknown although narrative says 'recovering'; Positive rechallenge recorded, but narrative says only that treatment was stopped	Reason for 'medically important' not stated; severity unknown.
SYN-PV-018	500 mg twice daily: Far above usual maximum (80 mg/day); entry error or overdose unknown	Mismatch: Atorvastatin recorded for anxiety	Positive rechallenge recorded, but narrative says only that treatment was stopped	Admission dates, reason and discharge status unknown.

Appendix A. Completed field guide

Field	Required for valid case	Student entry	Source evidence
Identifiable patient	Yes	Yes, met for all 18 cases. Age recorded in all cases (range 23-78); sex: Female 6, Male 5, Unknown 7. No names or initials (de-identified pack). Sex is unknown, not assumed.	case_scenarios.csv: Age, Sex
Identifiable reporter	Yes	Yes, met for all 18 cases (category only). Reporter type recorded: Consumer 4, Pharmacist 6, Physician 8. No name, qualification detail or country is supplied, so identifiability rests on the category in the training record.	case_scenarios.csv: Reporter
Suspect product	Yes	Yes, met for all 18 cases. 8 distinct products, each with dose and route (all oral). Batch, manufacturer and start date unknown.	case_scenarios.csv: SuspectProduct, Dose, Route
Adverse event	Yes	Yes, met for all 18 cases. 8 distinct reaction terms, one per case, as written in the source (not MedDRA-coded).	case_scenarios.csv: Reaction, Narrative
Chronology	No	Partly complete (optional field). Onset given as days after start in all cases (range 1-21). Calendar dates, stop date, event duration and concomitant start dates are unknown.	case_scenarios.csv: OnsetDays, Narrative
Seriousness	No	Complete as supplied (optional field). Hospitalisation 5, medically important 10, not serious 3. Severity not supplied (unknown). Seriousness is not severity.	case_scenarios.csv: Seriousness
Outcome	No	Complete from narrative (optional field). Recovered 6, recovering 6, not recovered 6. Outcome dates unknown. No fatal outcomes recorded.	case_scenarios.csv: Narrative

Appendix B. Limitations

- The cases are synthetic and several fields look machine-generated (for example indications that do not fit the drug). The analysis shows the method; it does not describe real drug behaviour.
- Causality categories are provisional. Without history, dates and verified doses, no case can be placed above Possible, and the five Conditional cases need their dechallenge contradictions resolved first.
- Time-to-onset, dose-range and risk-factor comments come from general pharmacological knowledge, not from a product label supplied in the pack. Expectedness is as supplied.
- Positive and negative rechallenge are recorded but undocumented, so they are treated as unverified and were not used to raise any category.
- Counts of drug-event pairs reflect the composition of the training pack only. No inference about frequency, risk or signal is made.
- This report is educational and is not clinical advice or a regulatory submission.